# Small stories around a bike

How a Bike Chain Works

Name _____ Date _____

Use numbers or a drawing to share your thinking.

**1.** Little joins 28 paper links, then adds 17 more. How many links are in the paper chain?

28 + 17 = _____

Show your thinking.

**2.** Big has 64 beads and threads 26 onto a model chain. How many beads are left?

64 − 26 = _____

Show your thinking.

**3.** Little records 23 slow pedal turns, then 19 more. How many pedal turns are recorded?

23 + 19 = _____

Show your thinking.

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**4.** A paper gear has 52 drawn teeth. Little colors 18. How many teeth still need color?

52 − 18 = _____

Show your thinking.

**5.** A bike has 32 teeth on its front sprocket and 16 on its back sprocket. How many teeth is that in all?

32 + 16 = _____

Show your thinking.

**6.** Little plans 15 gentle driveway laps and finishes 7. How many laps are left?

15 − 7 = _____

Show your thinking.

## A push has a path

The pedals were far from the back wheel. A loop of small metal links joined them. Little wondered how a push could travel.





## Bend at each pin

Tiny pins join the links. The chain bends at these joints without stretching like rubber. Sprocket teeth hook into the chain as it moves.





## Turning, then pulling

Pedals turn the front sprocket. Its teeth pull the chain. The chain pulls the back sprocket, which turns the back wheel.





## A loop on the table

Big turned a cardboard wheel. It pulled the bead loop, and the loop turned the smaller wheel. Little followed one bead around.





## One turn becomes two

The front had 32 teeth. The back had 16. One pedal turn carried enough chain past the back sprocket to turn it twice.





## A push goes with you

Little pedaled beside Big. The chain carried each pull to the back wheel. Where else could a loop carry motion?



## Make a little book



**1** Fold it into 8 boxes. Open it flat.



**2** Fold short end to short end. Cut the thick line. Open it.



**3** Fold the long way, pictures out. Push the ends in.




**4** Fold into a book. Page 1 goes on the front!





WONDER DAILY

# How a Bike Chain Works

# Give the chain a clear command

How a Bike Chain Works · Curiosity day

Name _____ Date _____

Choose a still view and tell what to count or draw.



Bend at each pin



Teeth to count



A loop on the table

Draw two sprockets joined by a chain. Add an arrow along the top of the chain.

chain · link · pin · sprocket · teeth · pedal · gear · pull · turn · wheel

Trace … name the loop you will follow

Count … name the teeth on a still sprocket

Draw … show how a loop joins two wheels

Does your command begin with an action word, such as Trace, Count, or Draw?